NEPAL PARICHAYA SERIES · VOLUME I

Nepal Parichaya: Physical Geography

Physiographic Divisions, River Basins, Climate Zones, and Biodiversity

Department of Information & Reference Compendium

National Reference Edition · 5 Pages

SECTION 1 · LOCATION & EXTENT

Geographic Coordinates and Dimensions

Stretching along the central arc of the Himalayas, Nepal bridges the Gangetic alluvial plains and the Tibetan plateau within less than two hundred kilometers.

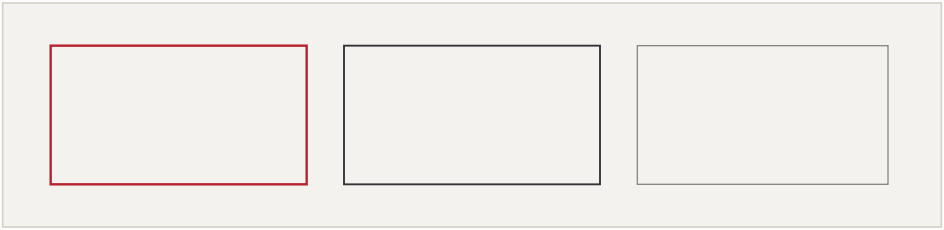
Nepal is a landlocked sovereign nation in South Asia situated between latitudes 26 deg 22 min N and 30 deg 27 min N, and longitudes 80 deg 04 min E and 88 deg 12 min E. The country covers a total land area of 147,516 square kilometers.

Its east-west length averages 885 kilometers from the Mechi River in the east to the Mahakali River in the west, while its north-south width varies between 145 kilometers and 241 kilometers, averaging 193 kilometers.

- Lowest elevation: Kechana Kalan (Jhapa District) at approximately 59 meters above sea level.
- Highest elevation: Sagarmatha (Mount Everest) at 8,848.86 meters above sea level.
- Share of global land area: 0.03% of the world and 0.3% of the Asian continent.

SECTION 2 · ECOLOGICAL BELTS

Himal, Pahad, and Terai Physiography



Reference Plate 2 — Structural breakdown and proportional distribution.

Topographically, Nepal rises in three distinct east-west ecological belts, each supporting specialized settlement patterns, agriculture, and ecosystems:

- Himalayan Region (Himal): Covers roughly 15% of total area above 3,000 meters. Home to eight of the worlds fourteen peaks above 8,000 meters—Sagarmatha, Kanchenjunga, Lhotse, Makalu, Cho Oyu, Dhaulagiri I, Manaslu, and Annapurna I.
- Hilly Region (Pahad): Encompasses roughly 68% of total land area between 600 and 3,000 meters, featuring the Mahabharat Lekh, Chure foothills, fertile tectonic valleys (Kathmandu and Pokhara), and terraced agriculture.
- Terai Region (Madhesh): Occupies roughly 17% of total land area in the southern alluvial belt below 600 meters, serving as the primary granary and industrial corridor of Nepal.

SECTION 3 · HYDROLOGY

Major River Systems and Glacial Basins

Over 6,000 rivers and rivulets traverse Nepal southward into the Ganges basin, categorized into first-grade snow-fed perennial systems, second-grade Mahabharat rivers, and third-grade seasonal Chure streams.

- Sapta Koshi Basin (East): Drains eastern Nepal via seven major tributaries—Tamur, Arun, Dudh Koshi, Likhu, Tama Koshi, Sun Koshi, and Indrawati.
- Sapta Gandaki / Narayani Basin (Central): Fed by Kali Gandaki, Trisuli, Budhi Gandaki, Marsyangdi, Daraundi, Seti, and Madi rivers.
- Karnali Basin (West): Longest river system inside Nepal (507 km), formed by Humla Karnali, Mugu Karnali, Tila, Seti, and Bheri.

SECTION 4 · CONSERVATION

Climatic Zones and Protected Areas

Because of extreme altitudinal variation, Nepal experiences five climatic zones within short horizontal distances: tropical/subtropical (<1,200m), warm temperate (1,200–2,100m), cool temperate (2,100–3,300m), alpine (3,300–5,000m), and tundra/nival (>5,000m).

Over 23 percent of Nepals territory is safeguarded across 12 National Parks, 1 Wildlife Reserve, 1 Hunting Reserve (Dhorpatan), and 6 Conservation Areas.